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1 The Escape Ridge of the Factor Skyline

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1.1 Abstract

Plotting the primes at their true Factor Skyline coordinates — cumulative x_{FS} against value — traces an “escape ridge” whose shape was first measured empirically by cubic and power regression ($y \approx 0.966 x^{0.86}$, $R^2 = 0.9993$). This note derives that shape from first principles. The horizontal coordinate grows as

$$x_{FS}(N) \sim 4e^{-\gamma} \frac{N^{3/2}}{\ln^2 N},$$

and the extra half-power over linear comes **entirely from the p^2 activation of the sieve sweeps** — a composite can only carry least prime factor p once $p^2 \leq n$, so the sweep sum is cut at $p \leq \sqrt{N}$. Inverting gives an escape ridge $y \sim X^{2/3}$ with a slow upward logarithmic drift, so the *local* power falls from ≈ 0.85 at $N \sim 10^3$ toward $2/3$ as $N \rightarrow \infty$. The empirical $x^{0.86}$ fit is a dead-on match at its scale, and the drift explains why a cubic outfit a single power law. The regression was not a curve-fit fishing for a number; it was measuring where the sieve turns its sweeps on.

1.2 1. The escape ridge

Recall the Factor Skyline coordinate: $x_{FS}(1) = 1$, and for $n \geq 2$,

$$x_{FS}(n) = x_{FS}(n-1) + dx(n), \quad dx(n) = \begin{cases} 1 & n \text{ prime} \\ \text{lpf}(n) & n \text{ composite.} \end{cases}$$

Every composite advances the horizontal coordinate by its least prime factor; every prime advances it by 1. The **escape ridge** is the locus of primes on this canvas: the points $(x_{FS}(p), p)$ for p prime. This is the object plotted in the “Factor Skyline 1000” Desmos sheet, whose table entries — e.g. $(x_{FS}, p) = (1496, 521)$, $(1499, 523)$ — the derivation below must reproduce (and does, exactly).

1.3 2. Growth of the horizontal coordinate

Summing the step sizes,

$$x_{FS}(N) \approx \pi(N) + \sum_{\substack{n \leq N \\ \text{composite}}} \text{lpf}(n).$$

The number of $n \leq N$ with least prime factor exactly p is $\approx N \cdot \frac{1}{p} \prod_{q < p} (1 - \frac{1}{q})$. A **composite** with $\text{lpf} = p$ requires $p^2 \leq n \leq N$, i.e. $p \leq \sqrt{N}$ — this is the sieve-of-Eratosthenes activation point, where prime p first begins crossing out numbers it is the smallest factor of. Each such composite contributes p , so

$$\sum_{\text{comp}} \text{lpf}(n) \approx N \sum_{p \leq \sqrt{N}} p \cdot \frac{1}{p} \prod_{q < p} \left(1 - \frac{1}{q}\right) = N \sum_{p \leq \sqrt{N}} \prod_{q < p} \left(1 - \frac{1}{q}\right).$$

By Mertens’ third theorem $\prod_{q < p} (1 - 1/q) \sim e^{-\gamma} / \ln p$, and by the prime number theorem $\sum_{p \leq y} 1/\ln p \sim y/\ln^2 y$. With $y = \sqrt{N}$,

$$\sum_{\text{comp}} \text{lpf}(n) \sim N \cdot e^{-\gamma} \cdot \frac{\sqrt{N}}{\ln^2 \sqrt{N}} = 4e^{-\gamma} \frac{N^{3/2}}{\ln^2 N}.$$

This dominates $\pi(N) \sim N/\ln N$, giving

$$x_{FS}(N) \sim 4e^{-\gamma} \frac{N^{3/2}}{\ln^2 N}, \quad 4e^{-\gamma} \approx 2.2458.$$

The $\sqrt{N}$ is the p^2 sweep. The half-power over linear is not incidental: it is the length of the sweep range $p \leq \sqrt{N}$. Without the p^2 activation wall the sum would run over all $p \leq N$ and the skyline would grow differently; cutting it at $\sqrt{N}$ is exactly what produces the $N^{3/2}$ ridge.

Numerical confirmation. The construction reproduces the Desmos table exactly ($x_{FS}(521) = 1496$, $x_{FS}(523) = 1499$). The local log-log slope of x_{FS} vs N climbs toward $3/2$:

N	$\sim 10^2$	$\sim 10^3$	$\sim 10^4$	$\sim 10^5$	$\sim 10^6$
exponent	1.18	1.17	1.24	1.29	1.33

and the constant ratio $x_{FS} \cdot \ln^2 N / N^{3/2}$ descends toward $4e^{-\gamma} = 2.25$ (from 4.9 at $N \sim 500$ to 3.2 at $N = 10^7$; the approach is slow, held back by the $\ln^2$ correction — see `graphics/FS_escape_ridge.png`, left panel).

1.4 3. The ridge and its drifting exponent

Inverting $X = x_{FS}(N) \sim C N^{3/2} / \ln^2 N$ for the prime value $y = N$ as a function of X :

$$y \sim \left(\frac{X}{C}\right)^{2/3} (\ln y)^{4/3}.$$

The leading behavior is $y \sim X^{2/3}$, with a slowly-growing $(\ln y)^{4/3}$ factor. Because that log factor inflates the *effective* exponent at finite scale, the **local** power b in $y \sim X^b$ is not $2/3$ until very large N ; it drifts down toward it:

range of N	up to 10^3	10^4	10^5	10^6	$\rightarrow \infty$
ridge power b	0.84–0.85	0.81	0.78	0.75	$2/3 \approx 0.667$

(see `graphics/FS_escape_ridge.png`, right panel).

1.5 4. The regression, explained

The “Factor Skyline 1000” sheet fit the ridge with a power law $y = 0.966 x^{0.86}$ and a cubic ($R^2 = 0.9993$). Both are now accounted for:

- **The power 0.86** matches the computed local ridge exponent ≈ 0.85 at $N \sim 10^3$ to within measurement — the fit was correct, and it was measuring the ridge at that scale.
- **Why the cubic fit better.** The true exponent is not constant; it slides from ≈ 0.85 toward $2/3$. A single power law can only match locally, whereas a cubic is a flexible local stand-in for a power law with a slowly-varying exponent — hence the near-perfect R^2 .

So the empirical regression was a faithful measurement of a real curve: the escape ridge $y \sim X^{2/3}$, reading $\approx X^{0.85}$ at the scale of a few thousand because of the logarithmic drift. And the curve’s bend is the signature of the sieve’s p^2 sweeps — the same activation-at-the-square that the wheel is built from.

1.6 Appendix. Reproduction

```
def spf_sieve(n):
    spf = list(range(n + 1)); i = 2
    while i * i <= n:
        if spf[i] == i:
            for j in range(i*i, n + 1, i):
                if spf[j] == j: spf[j] = i
            i += 1
    return spf

def factor_skyline(N):
    spf = spf_sieve(N)
    x = 1; ridge = [] # (x_FS, prime value)
    for n in range(2, N + 1):
        x += 1 if spf[n] == n else spf[n] # dx: 1 for primes, lpf for composites
        if spf[n] == n:
            ridge.append((x, n))
    return ridge

r = factor_skyline(1000)
print([X for X, p in r if p == 521]) # -> [1496], matching the Desmos table
```

1.7 References

- A. Proxmire, *The Architectural Foundation of the Factor Skyline* (this repository) — the x_{FS} coordinate and activation at prime squares.
- A. Proxmire, *Synthesis — Two Operations on the Factor Skyline* (this repository) — the wheel, the sweeps, and doubling.
- Mertens' third theorem; the prime number theorem.
- Figure: `graphics/FS_escape_ridge.png`.